Week 1 Questions

1. Identify which of the following are extensive properties and which are intensive properties: (a) a $10\,m^3$ volume, (b) $30\,J$ of kinetic energy, (c) a pressure of $90\,kPa$, (d) a stress of $1000\,kPa$, (e) a mass of 75 kg, and (f) a velocity of $60\,m/s$.
   Convert all extensive properties to intensive properties assuming $m=60\,kg$.
2. The temperature of a body is 50°F. Find its temperature in °C, K, and °R.
3. Express the following quantities in terms of base SI units (kg, m, and s): (a) power, (b) heat flux, and (c) specific weight.
4. Determine the weight of a mass of 12 kg at a location where the acceleration of gravity is 9.77 m/s^2.
5. Complete the following if $g=9.81\,m/s^2$ and $V=10^4\,L$. Keep two decimal places.

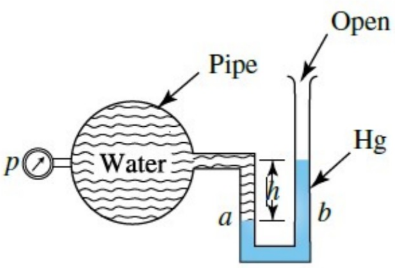
	$v(m^3/kg)$	$\rho(kg/m^3)$	$\gamma(N/m^3)$	$m(kg)$	$W(N)$
(a)	20.00				
(b)		2.00			
(c)			4.00		
(d)				100.00	
(e)					100.00

6. Complete the following if $p_{atm}=100\,kPa(\gamma_{Hg}=13.6\,\gamma_{water})$.

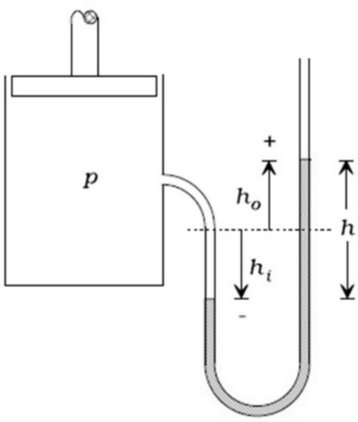
	kPa Gage	kPa abs	$mmHg$ abs	$m\,H_2O$
(a)	5			
(b)		150		
(c)			30	
(d)				30

7. Express the kinetic energy $\frac{1}{2}mV^2$ in acceptable terms if $m=12\,kg$, $V=4\,m/s$.
8. Calculate the force necessary to accelerate a 8000-kg rocket vertically upward at the rate of 28 m/s^2.
9. Express a pressure gage reading of 235 kPa in absolute pascals at an elevation of 1600 m (using p_{atm} at the given elevation in the textbook).

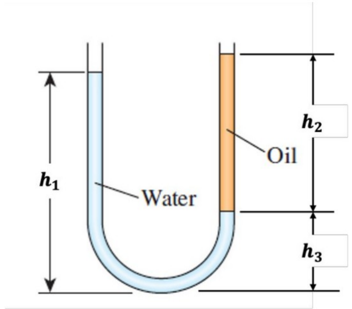
10. A 2200-kg automobile traveling at 90 kph hits the rear of a stationary, 1000-kg automobile. After the collision, the large automobile slows to 60 kph , and the smaller vehicle has a speed of 78 kph. What has been the increase in internal energy, taking both vehicles as the system?
11. Calculate the force due to the pressure acting on the 1.2-m diameter horizontal hatch of a submarine submerged 800 m below the surface.
12. If the mass of an object is 15 lbm, what is its weight, in lbf, at a location where g = 32.5 ft/s^2?
13. The mass of air in a room $3\,m \times 4\,m \times 25\,m$ is known to be $350\,kg$. Determine the density, specific volume, and specific weight.
14. A tube can be inserted into the top of a pipe transporting liquids, providing the pressure is relatively low, so that the liquid fills the tube a height h. Determine the pressure in a water pipe if the water seeks a level at height h = 152 cm above the centre of the pipe.
15. Determine the water pressure if the manometer reading is $0.45\,m$. Mercury is 13.6 times heavier than water.



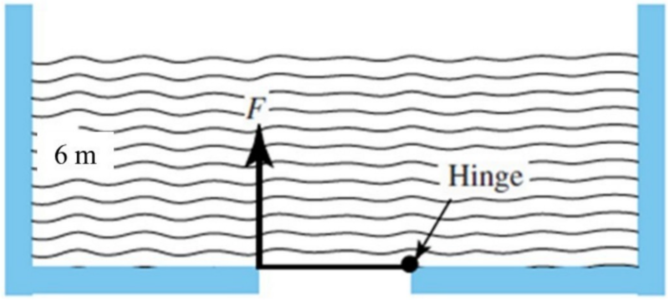
16. A mercury manometer connected to a tank of air reads h_o and h_i. What is the gauge pressure in MPa? Given $h_o = -h_i = 15\,cm$.



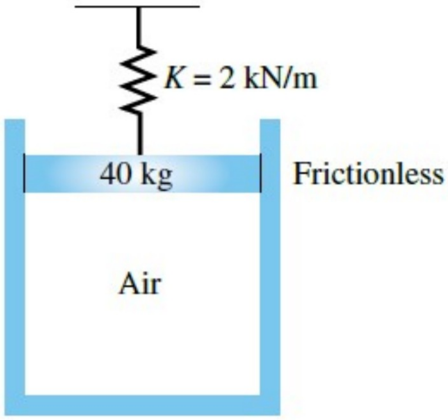
17. A U-tube with arms open to the atmosphere. Water is poured into the U-tube from one arm, and light oil from the other. One arm contains water, while the other arm contains both fluids. Determine the height of each fluid in that arm. Given: $\rho_{oil} = 790\,kg/m^3$, $h_1 = 50\,cm$ and $h_3 = \frac{1}{4} h_2$.



18. A horizontal 2.5-m-diameter gate is located in the bottom of a water tank. Determine the force F required to just open the gate.



19. Calculate the pressure in the 150-mm diameter cylinder. The spring is compressed 35 cm.



20. Determine the pressure difference between the water pipe and the oil pipe. The top of the mercury column is at the same elevation as the middle of the oil pipe.